



CS 59300

Application of Deep Learning

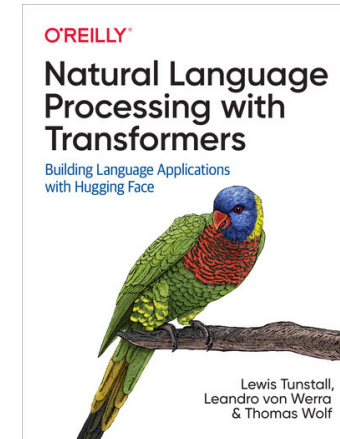
Hugging Face

Instructor: Dr. Zesheng Chen



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Reference



- <https://github.com/nlp-with-transformers/notebooks>
- <https://huggingface.co/transformersbook>

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Outline

- Introduction
- Pipelines
- Inference Providers

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Hugging Face

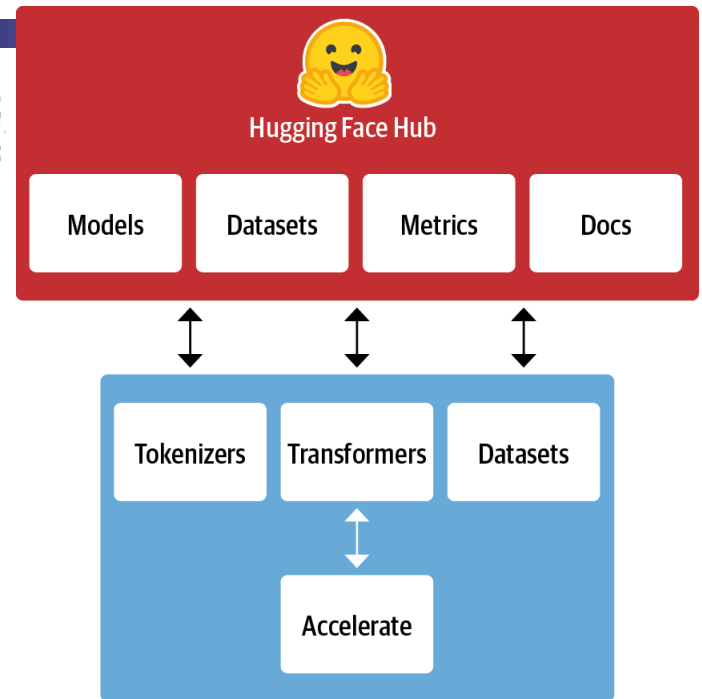
- Is called “**GitHub of AI**”
- Is a company and **open-source** community platform
- Makes AI and ML more **accessible**
- Is best known for its “**Transformers**” library
 - An open-source toolkit that provides easy access to **state-of-the-art** NLP models like BERT, GPT-2, and GPT-3
 - A **standard** in the field
 - <https://github.com/huggingface/transformers>

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Hugging Face History

- In 2016, Hugging Face was founded by French entrepreneurs Clément Delangue (CEO), Julien Chaumond (CTO), and Thomas Wolf (CSO).
- Hugging Face initially started as a **chatbot** application targeted at teenagers.
- A milestone is to create **Transformers** library at GitHub on 2018.
- In March 2021, Hugging Face raised US\$40 million in a Series B funding round.
- In August 2023, the company announced that it raised \$235 million in a Series D funding, at a \$4.5 billion valuation.

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Hugging Face Links

- <https://huggingface.co/>
- <https://huggingface.co/models>
- <https://huggingface.co/datasets>
- <https://huggingface.co/tasks>

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Outline

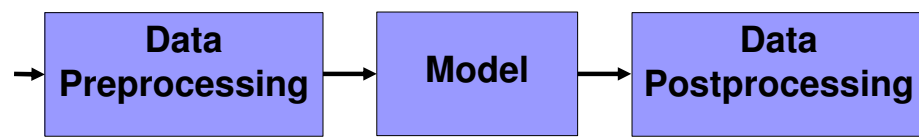
- Introduction
- **Pipelines**
- Inference Providers

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What are Pipelines For Hugging Face Transformer?

- Pipelines **abstract** away **all the steps** needed to convert raw text into a set of predictions from a fine-tuned model.



Examples

- Text Classification
- Named Entity Recognition
- Question Answering
- Summarization
- Translation
- Text Generation



Outline

- Introduction
- Pipelines
- **Inference Providers**



Inference Providers

- Inference Providers give developers access to hundreds of machine learning models, powered by **world-class** inference providers.
- They are also integrated into our client SDKs (for **JS and Python**), making it easy to explore **serverless inference** of models on your favorite providers.
 - Similar to ChatGPT API
- More information at <https://huggingface.co/docs/inference-providers/index>

Why Choose Inference Providers?

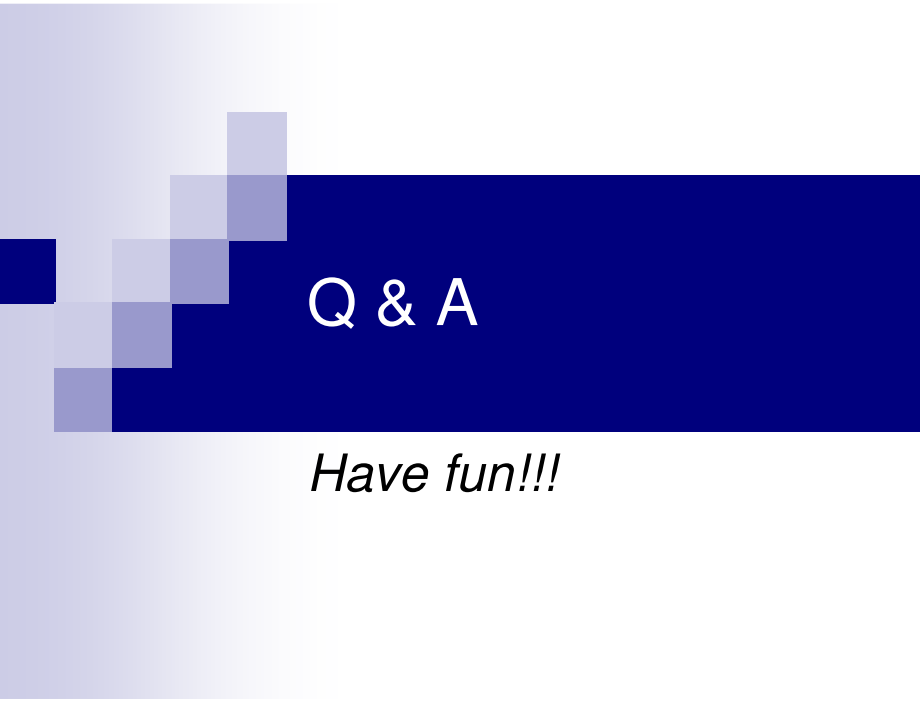
- Instant Access to Cutting-Edge Models
- Zero Vendor Lock-in
- Production-Ready Performance
 - Text Generation
 - Image and Video Generation
 - Search & Retrieval
 - Traditional ML Tasks
- Example: <https://huggingface.co/docs/inference-providers/guides/first-api-call>

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Summary

- Understand what are pipelines in Hugging Face Transformer library
- Apply Transformer pipelines to develop the application of deep learning (DL)
- Apply a model from inference providers

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Q & A

Have fun!!!